

Statistics Mat Working Towards Year 6

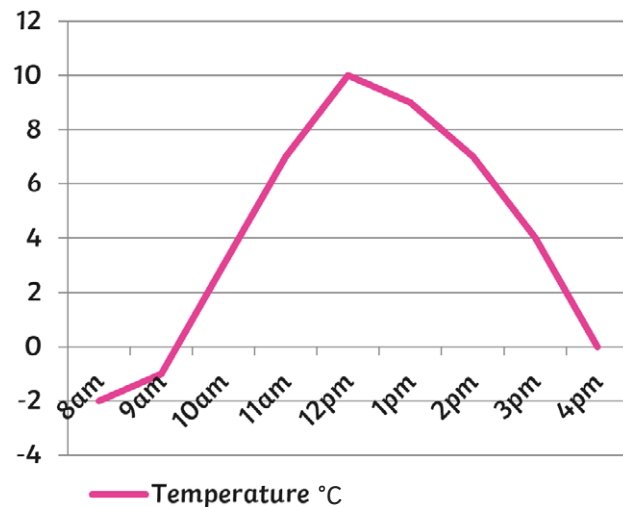
1

Line Graphs

a

Interpret line graphs and use these to solve problems.

Here is a line graph showing the temperature at different times in the day.



1. What is the highest temperature?

2. How many times is the temperature 6°C?

Mean

b

Calculate and interpret the mean as an average.

Here are some scores in a times tables test taken by 6 children. What is the mean of the results?

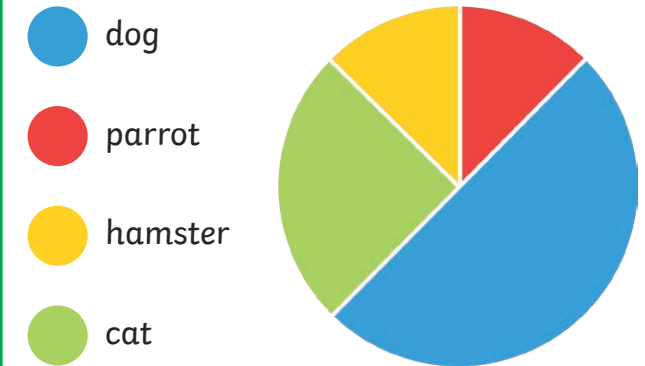
6	4	9	8	2	7
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Pie Charts

c

Interpret pie charts and use these to solve problems.

A group of 40 children were asked to name their favourite animal. Here is a pie chart showing the results:



This pie chart represents **40 children**.

1. How many children chose dogs as their favourite pet?

2. How many children chose hamsters as their favourite pet?

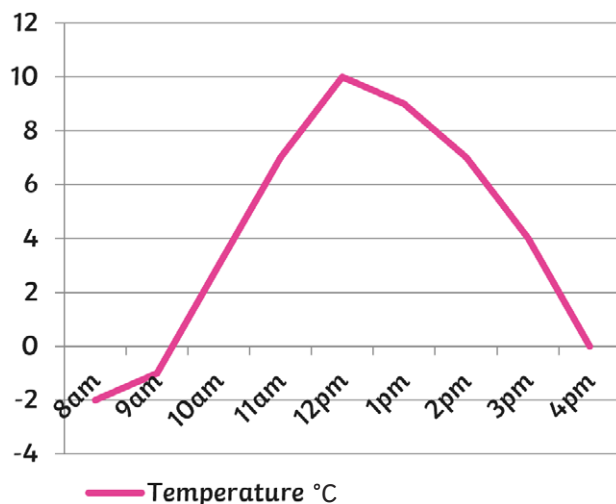
Statistics Mat Expected Year 6

1

Line Graphs

Interpret line graphs and use these to solve problems.

Here is a line graph showing the temperature at different times in the day.



- What is the highest temperature?

- How many times is the temperature 6°C?

Line Graphs

The temperature was recorded the following day. Construct a line graph from the data in this table.

Time	Temperature °C
9am	4
10am	8
11am	12
12pm	14
1pm	13
2pm	10
3pm	9
4pm	6

Mean

Calculate and interpret the mean as an average.

Here are some scores in a times tables test taken by 6 children. What is the mean of the results?

6	4	9	8	2	7
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Explain how to find the mean of a set of 5 numbers.

Pie Charts

Interpret pie charts and use these to solve problems.

A group of 40 children were asked to name their favourite animal. Here is a pie chart showing the results:

- dog
- parrot
- hamster
- cat



This pie chart represents **40 children**.

- How many children chose dogs as their favourite pet?

- How many children chose hamsters as their favourite pet?

A different group of children were asked to name their favourite animal. Here is a table showing the results. Construct a pie chart from the data:

Pet	Number of children
Dog	16
Cat	12
Rabbit	4

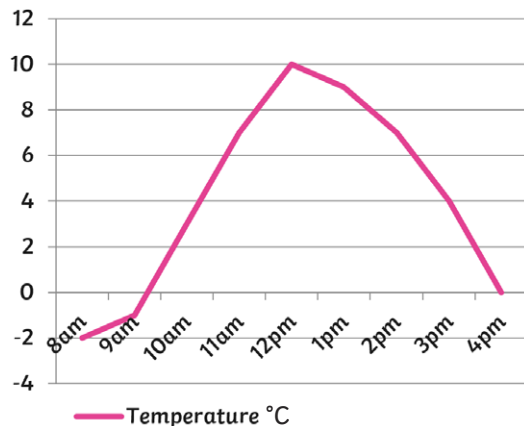
Statistics Mat Greater Depth Year 6

1

Line Graphs

Interpret line graphs and use these to solve problems.

Here is a line graph showing the temperature at different times in the day.



- What is the highest temperature?

- At which times is the temperature 6°C?

- Between which hours is the smallest change in temperature?

Record the temperature over a school day and use the data to construct a line graph.

Mean

Calculate and interpret the mean as an average.

Here are some scores in a times tables test taken by 6 children. What is the mean of the results?

6	4	9	8	2	7
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In another test, the same 6 children achieved a mean of 12. Here are 5 of the scores. Explain how to find what is the sixth score.

18	13	10	7	13	
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Pie Charts

Interpret pie charts and use these to solve problems.

A group of 40 children were asked to name their favourite animal. Here is a pie chart showing the results:

- dog
- parrot
- hamster
- cat



This pie chart represents **40 children**.

Explain how you can calculate that 5 people chose the parrot.

A different group of children were asked to name their favourite animal. Here is a table showing the results. Construct a pie chart from the data:

Pet	Number of children
Dog	10
Cat	12
Rabbit	4
Hamster	10

Statistics Mat Working Towards Year 6 **Answers**

1

Line Graphs

a

1. What is the highest temperature?
10°C
2. How many times is the temperature 6°C?
Twice

Mean

b

What is the mean of the results?
To find the mean of results, add all of them together
 $(6 + 4 + 9 + 8 + 2 + 7 = 36)$
and then divide the answer by 6
 $(36 \div 6 = 6)$.
The average score was 6.

Pie Charts

c

1. How many children chose dogs as their favourite pet?
20
2. How many children chose hamsters as their favourite pet?
5

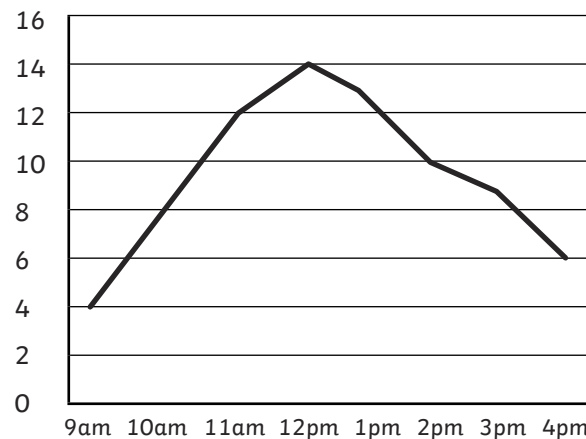
Line Graphs

a

- What is the highest temperature?
10°C
- How many times is the temperature 6°C?
Twice

Line Graphs

The temperature was recorded the following day. Construct a line graph from the data in this table.



Mean

b

What is the mean of the results?

To find the mean of results, add all of them together ($6 + 4 + 9 + 8 + 2 + 7 = 36$) and then divide the answer by 6 ($36 \div 6 = 6$).

The average score was 6.

To find the mean of 5 results, add all 5 results together and then divide the answer by 5. E.g. 7, 6, 3, 4, 5 Add these together makes a total of 25. $25 \div 5 = 5$. The mean is 5.

Pie Charts

c

- How many children chose dogs as their favourite pet?
20
- How many children chose hamsters as their favourite pet?
5

Line Graphs

a

- What is the highest temperature?
10°C
- At which times is the temperature 6°C?
11 am and 2pm
- Between which hours is the smallest change in temperature?
8am and 9am

Accept a line graph which shows temperatures over a school day.

Mean

b

What is the mean of the results?

The average score was 6.

To find the missing value, write out the calculation that you would use to find the mean.

$$18 + 13 + 10 + 7 + 13 + ? = 12$$

6

You then need to work out what number divided by 6 = 12. If you use the inverse, you know $6 \times 12 = 72$

The scores must add to make 72 as we know $72 \div 6 = 12$.

**The scores that we have add to 61 ($18 + 13 + 10 + 7 + 13$) so $72 - 61 = 11$
The missing score is 11.**

Check the calculation.

$$18 + 13 + 10 + 7 + 13 + 11 = 72.$$

$$72 \div 6 = 12$$

Pie Charts

c

The pie chart represents 40 children. Half of the pie chart is blue so 20 children chose dogs. A quarter of the pie chart is green. A quarter of 40 is 10. So 10 children chose cats. An eighth of the pie chart is red so and eighth of 40 is 5. 5 children chose parrot.

When we add all of the results together we can see that 36 children took part in the survey. Now you can show each result as a fraction: $\frac{10}{36}$, $\frac{12}{36}$, $\frac{4}{36}$, $\frac{12}{36}$. We know that there are 3600 in a full pie chart and we can change 36 into 360 by multiplying by 10, therefore if we change each fraction so that the denominator is out of 360, we will know how many degrees each segment is worth. $\frac{10}{36} = \frac{100}{360}$ so dog = 1000 and the same for hamsters. $\frac{12}{36} = \frac{120}{360}$ so cat = 1200 $\frac{4}{36} = \frac{120}{360}$ so rabbit = 400

Pie Chart to show favourite pets.